Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. **Let $A_1, A_2, \ldots, A_n$ be matrices such that the product $A_1A_2\cdots A_n$ is defined.**

   Express $(A_1A_2\cdots A_n)^T$ in terms of $A_1^T, A_2^T, \ldots, A_n^T$.

   Using the transpose property

   $$(AB)^T = B^T A^T,$$

   and applying it successively, we obtain:

   $$\begin{aligned}(A_1A_2\cdots A_n)^T &= ((A_1A_2\cdots A_{n-1})A_n)^T \\ &= A_n^T(A_1A_2\cdots A_{n-1})^T \\ &= A_n^T A_{n-1}^T(A_1A_2\cdots A_{n-2})^T \\ &\vdots \\ &= \boxed{A_n^T A_{n-1}^T\cdots A_2^T A_1^T}.\end{aligned}$$

   So the transpose of a matrix product is the product of the transposed matrices in reverse order.

2. Final item. Remove when done.